

Agentspan Course

A hands-on course workspace featuring **three progressively complex AI agents** built with the **Agentspan** framework · covering conversation memory, guardrails, and web research.

· Agents Overview

#	Agent	File	Description
1	Personal Assistant	agents/agent1.py	Conversational agent with time tool and memory
2	Customer Support	agents/agent2.py	Guardrailed agent with structured Pydantic responses
3	Research Agent	agents/agent3.py	Web research agent using Firecrawl for live searches

· Getting Started

Prerequisites

- Python 3.10+
- OpenAI API key
- Firecrawl API key (for Agent 3)

Installation

```
cd Agentspan-Course
uv sync
```

Environment Variables

Create a `.env` file:

```
OPENAI_API_KEY=sk-...
FIRECRAWL_API_KEY=...
```

Run Any Agent

```
uv run python agents/agent1.py
uv run python agents/agent2.py
uv run python agents/agent3.py
```

· Agent Details

Agent 1 · Personal Assistant (agents/agent1.py)

Capabilities:

- Current time retrieval tool
- Conversation memory across turns
- Friendly, helpful personality

Logic Flow:

- 1 User sends a message
 - 2 Agent checks if a tool call is needed (e.g., "What time is it?")
 - 3 Tool result is injected into context
 - 4 Agent responds with memory of previous messages
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Agent 2 · Customer Support (agents/agent2.py)

Capabilities:

- Input guardrails (blocks off-topic or harmful queries)
- Structured Pydantic response schema
- Professional support tone

Logic Flow:

- 1 User query passes through guardrail check
 - 2 If valid · Agent processes and responds with structured output
 - 3 If blocked · Guardrail rejection message returned
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Agent 3 · Research Agent (agents/agent3.py)

Capabilities:

- Live web search via Firecrawl API
- Multi-step research workflows
- Source-backed answers

Logic Flow:

- 1 User asks a research question
 - 2 Agent uses Firecrawl to search the web
 - 3 Results are analyzed and synthesized
 - 4 Answer is returned with sources
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· Project Structure

```
Agentspan-Course/  
... agents/  
...   ... agent1.py      # Personal assistant with memory  
...   ... agent2.py      # Customer support with guardrails  
...   ... agent3.py      # Research agent with Firecrawl  
... reports/             # Generated research reports  
... pyproject.toml        # Dependencies  
... uv.lock
```

· Dependencies

- agentspan

- firecrawl-py
 - pydantic
 - python-dotenv
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· License

Educational project · use freely for learning and reference.